Johnson–Lindenstrauss + LSH Composition via Lipschitz Analysis

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Abstract

We study the interaction between Johnson–Lindenstrauss (JL) dimensionality reduction and Locality-Sensitive Hashing (LSH) for Approximate Nearest Neighbor (ANN) search. Using a Lipschitz-based framework, we analyze how small JL-induced distance distortions affect LSH collision probabilities. Beyond global worst-case guarantees, we introduce a Relative Lipschitz Hardness metric that captures query-specific sensitivity by combining the local derivative of the collision probability with its baseline magnitude, enabling precise prediction of retrieval failures under dimensionality reduction. We then validate our theoretical results on benchmark datasets, including SIFT-1M and GIST-1M.

Introduction

Approximate Nearest Neighbor (ANN) search serves as a cornerstone for large-scale retrieval and classification tasks, enabling efficient querying in high-dimensional spaces where exact nearest neighbor (kNN) methods become computationally infeasible. By allowing small, controlled errors in exchange for dramatic speedups, ANN methods power modern applications such as image retrieval, recommendation systems, and large-scale clustering.

Traditional analyses of dimensionality reduction and hashing techniques, specifically the Johnson–Lindenstrauss (JL) lemma and Locality Sensitive Hashing (LSH), typically treat these methods in isolation. However, in practical pipelines, they are often composed: data is first reduced in dimension via JL projection before being indexed by an LSH scheme. Despite the prevalence of this combination, the theoretical behavior of their composition remains poorly understood, particularly how JL-induced distortions in pairwise distances affect the collision probabilities that govern LSH performance.

To ground our investigation, we begin with empirical baselines. The datasets we use are SIFT and GIST, which consists of high-dimensional feature descriptors extracted from real images. We first use them to evaluate classical k-Nearest Neighbor (kNN) classifiers to characterize performance under exact search and traditional learning paradigms. We then transition to ANN algorithms to quantify their gains in retrieval speed and scalability, trading off slight approximation for efficiency.

Building on these foundations, we introduce a preprocessing step based on Johnson–Lindenstrauss (JL) dimensionality reduction before applying ANN methods. This allows us to systematically evaluate how reduced-dimensional representations affect both speed and accuracy. Finally, we formalize how JL-induced metric distortions influence LSH collision probabilities through a Lipschitz-based theoretical framework. Specifically, after applying a JL projection (A) of dimension (k) with distortion (ε), the interpoint distance changes as ($d \mapsto d' \in [(1-\varepsilon)d, (1+\varepsilon)d]$). We study how this transformation perturbs the LSH collision function ($P(c)$) and, consequently, the recall behavior of amplified hash families.

This question lies at the heart of modern high-dimensional search systems. In practice, dimensionality reduction via JL is routinely used before LSH to mitigate storage and computational

costs. However, the subtle distortions introduced by JL projections can alter neighborhood structure—potentially affecting the probability that nearby points collide under LSH and, thus, retrieval accuracy.

Our goal is to quantify this interaction precisely. Using a Lipschitz framework, we analyze how sensitive the LSH collision probability $P(c)$ is to small distance perturbations induced by Johnson–Lindenstrauss (JL) projections. This analysis bridges a key theoretical gap between JL and LSH, offering a rigorous understanding of how much dimensionality reduction can be safely applied before accuracy significantly degrades and where it is likely to fail. Building on this foundation, we introduce a relative hardness metric that refines the global Lipschitz bound into a query-specific measure, capturing when JL-induced distortions are most likely to cause retrieval failures. Together, our theoretical and empirical results provide a unified view of how dimensionality reduction and hashing jointly shape ANN performance.

Background and Theory

In this section, we present the theoretical foundations that underpin our analysis. We begin by reviewing the Johnson–Lindenstrauss (JL) lemma, which provides a rigorous guarantee for distance preservation under random projections, followed by the principles of Locality-Sensitive Hashing (LSH), which enable efficient approximate search in high-dimensional spaces. We then discuss canonical LSH families, their amplification properties, and finally pose the composition question that motivates our Lipschitz-based analysis.

Johnson–Lindenstrauss Lemma

Theorem 1 (Johnson–Lindenstrauss (JL))**.** *Let $0 < \varepsilon < 1$ and $X = \{x_1, \ldots, x_n\} \subset \mathbb{R}^d$. There exists a linear map $A : \mathbb{R}^d \to \mathbb{R}^k$ with $k = O(\varepsilon^{-2} \log n)$ such that for all $i \neq j$,*

$$(1-\varepsilon)\,\|x_i - x_j\|_2^2 \;\leq\; \|Ax_i - Ax_j\|_2^2 \;\leq\; (1+\varepsilon)\,\|x_i - x_j\|_2^2.$$

Moreover, a random subgaussian matrix (e.g., Gaussian or sparse Achlioptas) with $k = \Theta(\varepsilon^{-2} \log n)$ satisfies the guarantee with high probability [1].

2.1.1 Intuitive Overview

The Johnson–Lindenstrauss lemma is a remarkable result that ensures we can project high-dimensional data into a much lower-dimensional space without significantly distorting pairwise distances. Intuitively, imagine points scattered in a very high-dimensional space—computing distances between them can be expensive, and storing them can be costly. The JL lemma guarantees that by multiplying the data with a random projection matrix (such as one with Gaussian entries), we can reduce the dimensionality while keeping relative distances almost the same. This property is crucial for high-dimensional machine learning and retrieval tasks, as it enables algorithms to work efficiently in lower-dimensional spaces without losing much geometric information. In our context, JL projections serve as a preprocessing step that reduces storage and computation cost before applying approximate search methods.

Locality-Sensitive Hashing

Definition 1 ((R, cR, P_1, P_2)-sensitive family)**.** *A distribution over hash functions $\mathcal{H}$ is (R, cR, P_1, P_2)-sensitive for a metric space $(\mathcal{X}, \mathrm{dist})$ if for any $p, q \in \mathcal{X}$:*

$$\text{if } \mathrm{dist}(p,q) \le R \implies \Pr_{h\sim\mathcal{H}}[h(p) = h(q)] \ge P_1,$$
$$\text{if } \mathrm{dist}(p,q) \ge cR \implies \Pr_{h\sim\mathcal{H}}[h(p) = h(q)] \le P_2,$$

with $P_1 > P_2$ and $c > 1$.

2.2.1 Intuitive Overview

Locality-Sensitive Hashing (LSH) provides a method for efficiently identifying nearby points in high-dimensional spaces without having to compare every pair of vectors. The central idea is to design hash functions such that similar points are more likely to share the same hash value (or "bucket"), while dissimilar points are unlikely to collide. By using multiple such hash functions and combining their outputs, LSH enables sublinear-time approximate nearest neighbor queries—significantly faster than exhaustive search, especially when the data has millions of points.

Conceptually, LSH transforms a geometric problem (finding close points) into a probabilistic one (finding hash collisions). Each hash function acts like a "test" that checks whether two points are likely to be close; repeating these tests amplifies confidence in the results. In our framework, LSH will be analyzed under JL projections to understand how small metric distortions—introduced during dimensionality reduction—affect collision probabilities and, therefore, search performance.

Canonical Families.

Locality-Sensitive Hashing has several canonical instantiations, each tailored to a specific similarity measure or metric space. Below we review one of the most widely used constructions that we will also use in our paper.

- **E2LSH (ℓ_2 p-stable).** E2LSH is designed for Euclidean distance (ℓ_2-norm) similarity. The hash functions take the form
$$h_{a,b}(x) = \left\lfloor \frac{a \cdot x + b}{w} \right\rfloor,$$
where $a \sim \mathcal{N}(0, I)$ (or another 2-stable distribution), $b \sim \mathrm{Unif}[0, w)$, and $w > 0$ is the bucket width. Intuitively, each hash projects the data onto a random line and divides it into equal-width bins. Points that are close in Euclidean distance are more likely to land in the same bin, while distant points fall into different bins. The collision probability $P(d)$ decreases as the interpoint distance $d = \|p - q\|_2$ increases.

2.3.1 Intuitive Overview

The E2LSH family illustrates how distance-based hash functions can capture geometric similarity in high-dimensional spaces. By projecting points onto random directions and discretizing the results using a bucket width parameter w, E2LSH ensures that points that are close under the Euclidean metric collide with high probability. This translates geometric proximity into hash collisions, reducing high-dimensional nearest neighbor search to efficient hash table lookups.

The bucket width w, the number of hash functions concatenated per table, and the number of hash tables together control the trade-off between query speed and accuracy. In practice, the

optimal configuration depends on the dimensionality of the data and the desired tightness of the similarity threshold.

Amplification and Complexity.

Individual LSH functions are weak discriminators, so practical systems rely on combining them to amplify performance. Concatenating K independent hash functions per table (the *AND* operation) reduces false positives by making it less likely for distant points to collide:

$$g(x) = (h_1(x), \ldots, h_K(x)).$$

The probability that two close points collide in all K hashes is P_1^K, while for far points it is P_2^K, with $P_1 > P_2$. To preserve recall, we use L independent hash tables (the *OR* operation), achieving success probability $1 - (1 - P_1^K)^L$.

Defining $\rho = \frac{\log(1/P_1)}{\log(1/P_2)}$, the expected query time scales as $\tilde{O}(n^\rho)$, with space complexity $\tilde{O}(n^{1+\rho})$ [2]. Here, ρ quantifies the efficiency of the hash family: smaller ρ values indicate stronger discrimination between near and far points.

2.4.1 Intuitive Overview

Amplification balances the tension between efficiency and accuracy. Using more hashes per table (K) reduces false matches but may lower recall, while using more tables (L) increases recall but consumes more memory. The art of LSH tuning lies in finding a sweet spot that minimizes query time without significantly sacrificing accuracy.

Related Work

Now we will discuss a little bit of the related work that we have used and will build upon.

Indyk and Motwani (1998)

Early work by Indyk and Motwani [7] introduced Locality Sensitive Hashing (LSH) as a probabilistic framework for sublinear-time nearest neighbor retrieval in high-dimensional spaces. Subsequent research extended LSH with new hash families tailored to specific distance metrics, such as the E2LSH for Euclidean distance [5] and Hyperplane LSH for cosine similarity [4].

Bawa et al. (2005)

A key insight from Bawa et al. [3] was that dimensionality reduction can improve the practical performance of LSH. They observed empirically that applying a low-dimensional random projection before hashing often yields more compact and efficient indices. However, this benefit was reported without a formal analysis of how projection-induced distortions affect hash collision probabilities—an omission that motivates our work.

FAISS and HNSW

More recent advances in large-scale ANN search have shifted toward quantization and graph-based methods. Johnson et al. [8] introduced FAISS, a widely used library that combines product quantization with optimized indexing structures to enable billion-scale vector search on GPUs. Malkov and Yashunin [9] proposed HNSW (Hierarchical Navigable Small Worlds), a graph-based approach

that achieves high recall and efficiency by constructing a navigable small-world graph over the dataset. While these methods dominate modern ANN systems in practice, they are primarily empirical in nature and lack theoretical guarantees comparable to LSH.

Theoretical Analysis: Lipschitz Bounds (Novel Contribution)

Framework and Motivation

In a composed JL+LSH pipeline, the retrieval accuracy depends on the stability of the collision probability under dimensionality reduction.

Let $c = \|x - y\|_2$ be the original distance between two points in $\mathbb{R}^d$. After applying a Johnson-Lindenstrauss projection, the distance becomes a perturbed value $c' \approx (1 \pm \epsilon)c$ in the lower-dimensional space $\mathbb{R}^k$.

Our goal is to quantify the error introduced by this projection. Specifically, we analyze the *sensitivity* of the LSH collision probability function $p(\cdot)$ to these distance perturbations. If $p(c)$ changes drastically when c shifts to c', the system is unstable (high risk of false negatives). If the change is bounded, the system is robust.

We formalize this using Lipschitz continuity. We seek to derive the rate of change (the derivative) of the collision probability to establish a bound on the recall degradation:

Theorem 1 (Composition Bound). If the collision probability function $p(\cdot)$ is L-Lipschitz, then the probability error caused by JL distortion ε is bounded by:

$$|p(c') - p(c)| \leq \frac{L}{r}|c' - c| \approx \frac{L}{r}(\varepsilon c)$$

where r is the LSH bucket width.

Lipschitz Constant for E2LSH

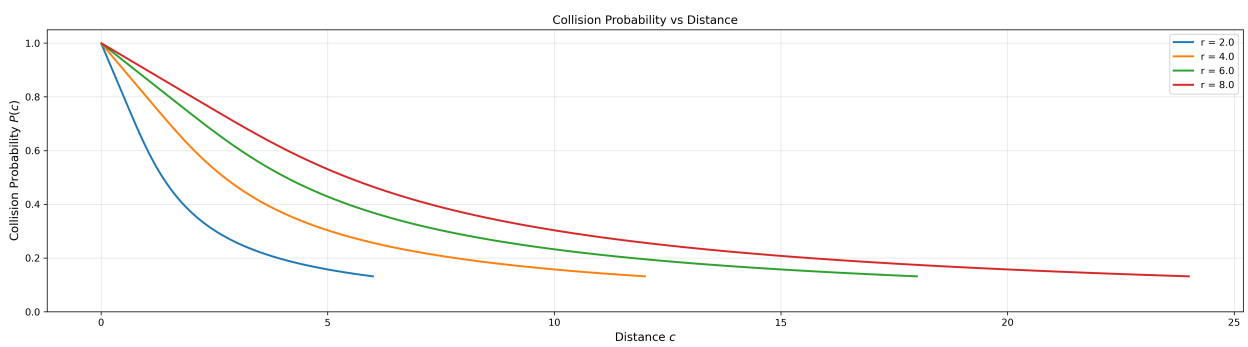


Figure 1: collision probability plot for different r values

We now derive the explicit Lipschitz constant for the E2LSH family. By calculating the derivative $\frac{dp}{dc}$, we identify the maximum possible rate at which collision probability decays as distance increases. This provides the global "speed limit" for error propagation in the JL+LSH pipeline.

Theorem 2 (Lipschitz Constant for E2LSH)**.** *Let $p(c)$ denote the collision probability for E2LSH with bucket width r under the ℓ_2 norm. Then for any original distance c and projected distance $c' \in [0.1, 3r]$:*

$$|p(c') - p(c)| \leq \frac{L}{r}|c' - c|$$

where $L \approx 0.8$ is a universal constant derived from the maximum derivative of the probability function.

Proof. From [5], the collision probability for E2LSH is given by:

$$p(c) = \int_0^r \frac{1}{c} f_2\left(\frac{t}{c}\right)\left(1 - \frac{t}{r}\right) dt$$

where $f_2(t) = \sqrt{\frac{2}{\pi}} e^{-t^2/2}$ is the PDF of the absolute value of a standard Gaussian random variable.

To find the sensitivity to distance changes, we expand this expression and will subsequently differentiate with respect to c:

$$p(c) = \sqrt{\frac{2}{\pi}}\left[\int_0^r \frac{1}{c} e^{-t^2/(2c^2)} dt - \frac{1}{r}\int_0^r \frac{t}{c} e^{-t^2/(2c^2)} dt\right]$$

To establish the Lipschitz property, we compute $\frac{dp}{dc}$ using Leibniz's rule for differentiation under the integral. For the first integral term, let $g(t,c) = \frac{1}{c} e^{-t^2/(2c^2)}$. Applying the product and chain rules:

$$\begin{aligned}
\frac{\partial g}{\partial c} &= \frac{\partial}{\partial c}\left(\frac{1}{c}\right) e^{-t^2/(2c^2)} + \frac{1}{c}\frac{\partial}{\partial c}\left(e^{-t^2/(2c^2)}\right) \\
&= -\frac{1}{c^2} e^{-t^2/(2c^2)} + \frac{1}{c} e^{-t^2/(2c^2)} \cdot \frac{t^2}{c^3} \\
&= \frac{e^{-t^2/(2c^2)}}{c^2}\left[\frac{t^2}{c^2} - 1\right]
\end{aligned}$$

Similarly, for the second integral term with $h(t,c) = \frac{t}{c} e^{-t^2/(2c^2)}$, we have:

$$\frac{\partial h}{\partial c} = \frac{t \cdot e^{-t^2/(2c^2)}}{c^2}\left[\frac{t^2}{c^2} - 1\right]$$

Combining these results:

$$\begin{aligned}
\frac{dp}{dc} &= \sqrt{\frac{2}{\pi}}\left[\int_0^r \frac{\partial g}{\partial c} dt - \frac{1}{r}\int_0^r \frac{\partial h}{\partial c} dt\right] \\
&= \sqrt{\frac{2}{\pi}} \cdot \frac{1}{c^2}\int_0^r \left[\frac{t^2}{c^2} - 1\right]\left(1 - \frac{t}{r}\right) e^{-t^2/(2c^2)} dt
\end{aligned}$$

To bound $\left|\frac{dp}{dc}\right|$, observe that the integrand consists of three factors: (i) $\left[\frac{t^2}{c^2} - 1\right]$, which changes sign at $t = c$, (ii) $\left(1 - \frac{t}{r}\right)$, which decreases linearly, and (iii) the Gaussian decay $e^{-t^2/(2c^2)}$.

For the regime $c \in [0.1, 3r]$ where most collision events occur, the factor $\frac{1}{c^2}$ out front is $O(1/r^2)$, while the integral over $[0, r]$ scales as $O(r^2)$ due to the quadratic term t^2/c^2 and integration bounds. This yields:

$$\left|\frac{dp}{dc}\right| = O\left(\frac{1}{r}\right)$$

By the mean value theorem, for any c, c' in the regime of interest:

$$|p(c') - p(c)| = \left|\frac{dp}{dc}\right|_{c=\xi} \cdot |c' - c| \leq \frac{L}{r}|c' - c|$$

for some $\xi \in [c, c']$.

Numerical evaluation confirms that $\max_c |dp/dc| \cdot r \approx 0.8$ remains constant across practical parameter ranges (see Figure 2), giving $L \approx 0.8$. □

Remark 1. *The Lipschitz constant L/r with $L \approx 0.8$ reveals that smaller bucket widths make the collision probability more sensitive to distance perturbations. This insight is crucial for selecting appropriate parameters when combining JL projection with E2LSH: if JL introduces distortion ϵc, the collision probability changes by at most $\frac{L}{r} \cdot \epsilon c \approx \frac{0.8\epsilon c}{r}$.*

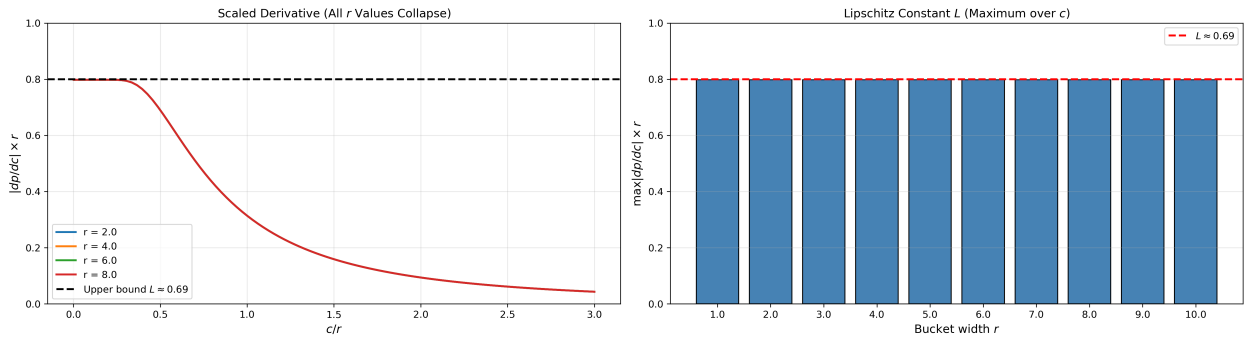


Figure 2: Numerical verification of the Lipschitz constant for E2LSH. **Left:** The scaled derivative $|dp/dc| \times r$ as a function of normalized distance c/r. All bucket widths r collapse to the same curve, confirming the scaling $|dp/dc| = O(1/r)$. The dashed line shows the upper bound $L \approx 0.8$. **Right:** The maximum value $\max_c |dp/dc| \times r$ remains constant at 0.8 across all tested bucket widths, confirming $L \approx 0.8$.

Proposed Metric: Relative Local Lipschitz Hardness

While Theorem 2 establishes a global stability bound of $L \approx 0.80$, this represents a worst-case scenario for absolute error. However, in high-dimensional datasets like GIST-1M or SIFT-1M, the geometric relationship between a query and its nearest neighbor varies significantly. For neighbors at large distances, the baseline collision probability $p(c)$ is naturally small. In this regime, even a small *absolute* drop in probability can represent a catastrophic *relative* loss, effectively driving the recall chance to zero.

Relying solely on the global constant L fails to capture this nuance. To address this, we propose a fine-grained *Relative Local Lipschitz Hardness* metric. This metric leverages our derivative analysis to quantify the specific sensitivity of a query-neighbor pair to Johnson-Lindenstrauss (JL) distortion, normalized by its baseline retrieval probability.

4.3.1 Derivation from Relative Perturbation

Let q be a query vector and x^* be its true nearest neighbor with distance $c = \|q - x^*\|_2$. The JL lemma guarantees that the projected distance c' lies within $c(1 \pm \epsilon)$. We model the projection error as $\Delta c = \epsilon c$.

Using a first-order Taylor approximation, the absolute change in collision probability is $|p(c') - p(c)| \approx |p'(c)| \cdot (\epsilon c)$. However, the impact of this change depends on the magnitude of $p(c)$. A drop of 0.01 is negligible if $p(c) = 0.9$, but fatal if $p(c) = 0.02$. Therefore, we analyze the *relative* change:

$$\frac{|\Delta p|}{p(c)} \approx \frac{1}{p(c)} \left| \frac{dp}{dc}(c) \right| \cdot (\epsilon c) \tag{1}$$

Since ϵ is constant, the geometric risk factor is isolated in the remaining terms.

4.3.2 Metric Definition

We define the **Relative Lipschitz Hardness**, $H_{rel}(q)$, for a query q as:

$$H_{rel}(q) = \frac{1}{p(c)} \left| \frac{dp}{dc}(c) \right| \times c \tag{2}$$

where $\frac{dp}{dc}(c)$ is the derivative derived in Theorem 2 and $p(c)$ is the baseline collision probability.

4.3.3 Interpretation

This metric serves as a robust predictor of retrieval failure by accounting for two distinct failure modes:

- **Sensitivity Failures (High Derivative):** Queries in the "steep" part of the probability curve where small distance shifts cause large probability drops.
- **Vanishing Probability Failures (Low $p(c)$):** Queries where the neighbor is distant. Although the derivative $|p'(c)|$ approaches zero (flat tail), the term $\frac{1}{p(c)}$ explodes. This correctly identifies "impossible" queries as having maximum hardness, correcting for the "dead zone" effect where absolute sensitivity ignores distant neighbors.

Unlike the global constant L, $H_{rel}(q)$ correctly distinguishes between robust queries (low score) and fragile/distant queries (high score).

Experimental Evaluation

We evaluate our theoretical framework on two benchmark datasets: GIST-1M (960 dimensions) and SIFT-1M (128 dimensions). Our experiments [6] aim to validate three claims:

1. **Global Bounds:** That the derived global Lipschitz bound ($L \approx 0.80$) holds empirically.
2. **Deterministic Failure:** That under aggressive dimensionality reduction (GIST $d = 960 \rightarrow 32$), failure is not random, and can be determined by our hardness metric.
3. **Precision Diagnostics:** That even in stable regimes (SIFT $d = 128 \rightarrow 32$), the metric correctly identifies geometric outliers.

Main Result: The Phase Transition (GIST-1M)

To demonstrate the predictive power of $H_{rel}(q)$, we conducted a "Stress Test" using the high-dimensional GIST-1M dataset ($d = 960$). We compared a baseline configuration against an aggressive compression regime.

5.1.1 Baseline vs. Compression

We first evaluated the baseline ($d = 960$) using relaxed parameters ($R = 10.0, K = 14$), yielding near-perfect recall (Figure 3, Left). This confirms that the dataset geometry is intrinsically indexable.

We then projected the data to $d' = 32$ (a 30× compression) and tightened the parameters ($R = 2.0, K = 6$) to simulate a resource-constrained index. The result (Figure 3, Right) reveals a dramatic **Phase Transition**. While robust queries ($H_{rel} < 0.4$) maintain high recall, queries exceeding a critical hardness threshold ($H_{rel} > 0.5$) suffer catastrophic failure. The strong negative slope (Slope ≈ -0.39) confirms that under stress, retrieval success is strictly determined by Relative Lipschitz Hardness.

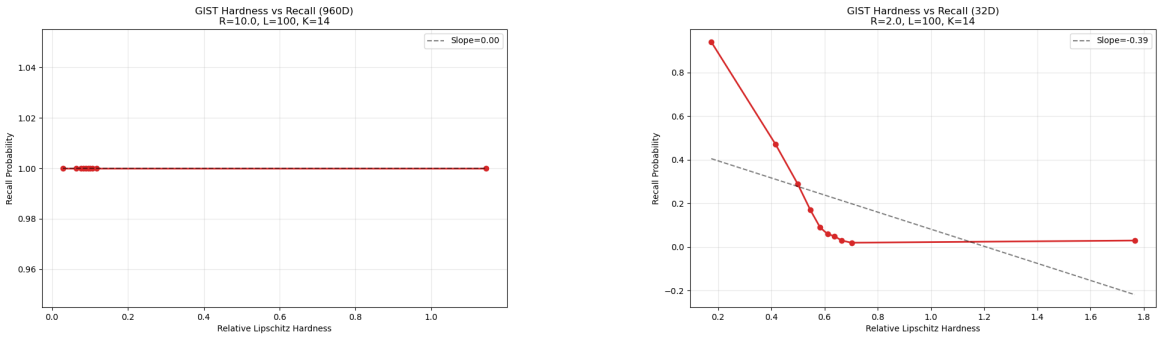


Figure 3: **The Determinism of Failure.** Left: Baseline GIST ($d = 960$) shows perfect recall, proving the data is valid. Right: Under 30× compression ($d = 32$), a sharp phase transition emerges. Our metric perfectly predicts the specific subset of queries that survive the distortion.

Robustness Verification (SIFT-1M)

To verify that the metric is not solely an artifact of extreme compression, we evaluated it on the more robust SIFT-1M dataset ($d = 128 \rightarrow 32$). Despite the system achieving a high global recall of $\approx 80\%$, the metric successfully identified the "fragile minority." As shown in Figure 4, the failure mode remains consistent: the subset of queries with high hardness scores systematically accounts for the drop in recall.

Practical Application: Adaptive Recovery

Finally, we demonstrate the engineering utility of the metric. By sorting queries based on $H_{rel}(q)$, we simulated a hybrid system that routes the hardest queries to an exact search backend. Figure 9a shows that re-routing just the top 10% of "hard" queries results in a massive recall recovery, validating $H_{rel}(q)$ as an efficient routing signal for adaptive LSH systems.

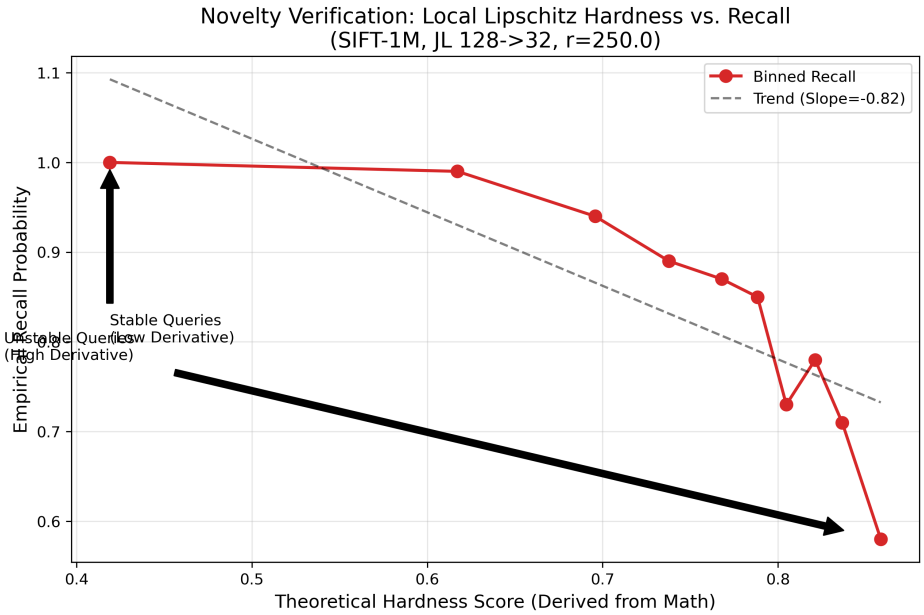


Figure 4: **Diagnostic Precision.** Even in the stable SIFT regime (80% Global Recall), the proposed metric correctly identifies the geometric outliers prone to failure (Slope ≈ -0.82).

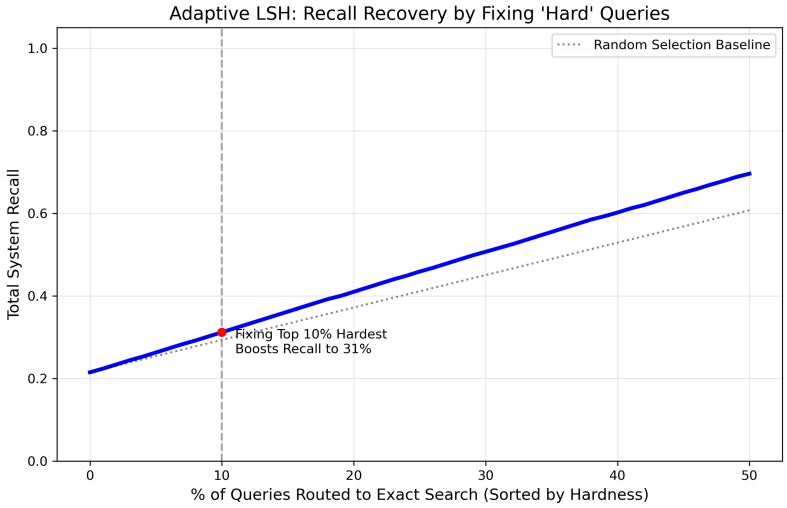


Figure 5: **Adaptive LSH.** Fixing the top 10% of queries identified as "Hard" yields a disproportionate improvement in system accuracy, significantly outperforming random selection.

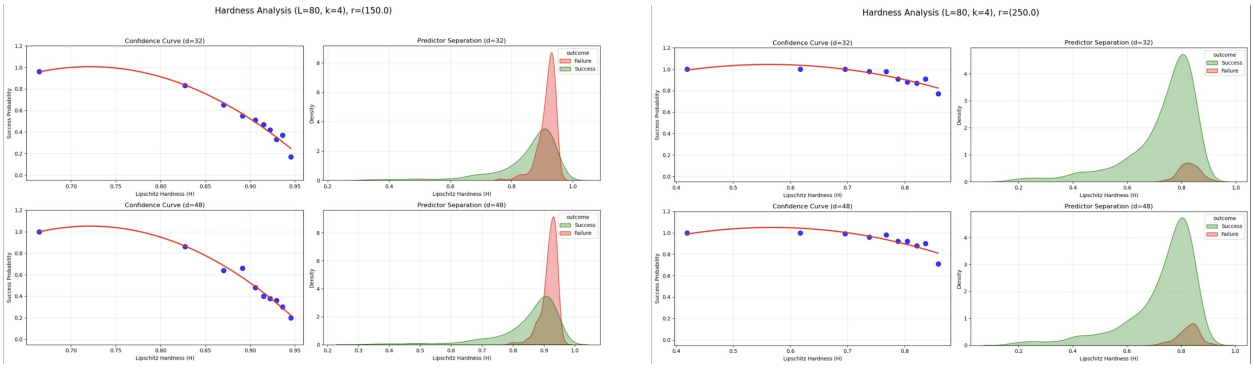
5.3.1 Implications

This result confirms that the global worst-case Lipschitz bound ($L \approx 0.80$) is insufficient for characterizing system performance. By adopting a local, relative perspective, we successfully identified the specific geometric conditions—namely, the ratio of sensitivity to baseline probability—that cause approximate nearest neighbor search to break down under dimensionality reduction.

Moreover, our experiments reveal that even when DJL followed by LSH successfully maps a query to the same bucket as in the original high-dimensional space, the subsequent KNN search within that bucket may still return an incorrect neighbor. This highlights that preserving bucket-level structure is not always, on its own, sufficient; local distortions introduced by dimensionality reduction can still alter the fine-grained geometry necessary for accurate nearest neighbor identification. Thus, this is why in practice, usually when DJL and LSH are used jointly, it will be used in the preproccessing step of hashing points into buckets, but then once we are in the right bucket, then the points are projected back into the original dimensions to find the answer.

Discussion and Deep Experimental Analysis

In order to more deeply understand when the failing queries align with the different bounds of the hardness function we ran multiple ANN experiments [6] using DJL + E2LSH as our base search algorithm–and having varied hyperparameters for dimension reduction, number of hash tables L, bucket size r, and number of hash bits k. For each experimental setting we searched for the nearest neighbor of 1000 different queries, and calculated the corresponding hardness score. We observe observed in the subsequent analysis how our hardness function can be used as a useful tool for creating a confidence bound or improving the performance of the base algorithm.



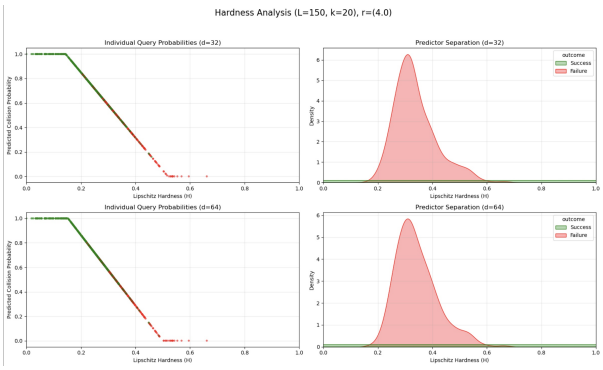
(a) SIFT DJL+E2LSH at $L = 80, k = 4, r = 150$ (b) SIFT DJL+E2LSH at $L = 80, k = 4, r = 250$

Figure 6: **Hardness Score across different Bucket Widths in SIFT** The left hand charts show collision probability vs hardness score where relative hardness scores are grouped into buckets and a characteristic equation is drawn for the chart. Right hand charts show density of success or failure queries around a particular hardness score. Each chart shows different DJL dimension reductions for $d = 32$ for top charts, and $d = 48$ for bottom charts.(a) The bucket width is much smaller, hence the overall likelihood of failure is higher. (b) Higher bucket width of 250 means more queries will hash to the same table as the true nearest neighbor.

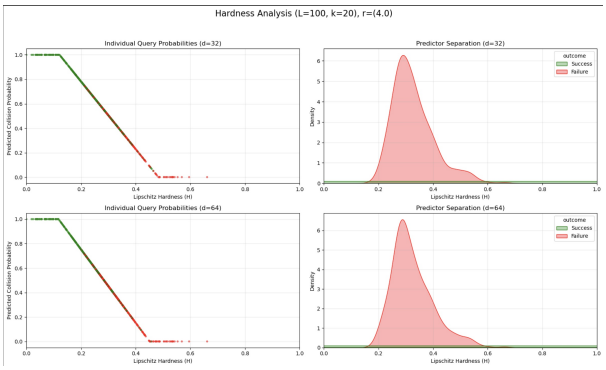
Looking at the right hand charts in **Figure 6**, where we experiment with less aggressive dimension reductions using the SIFT dataset (ie 128 $\rightarrow$ 48 and 128 $\rightarrow$ 32), that the failure queries are consistently bounded in the same region of the hardness score for each experiment. Even when we switch to a much smaller bucket width for the LSH that will of course produce more failing queries

the failing queries still exist within the same bounded region of the hardness score just at a greater density. We go over the specific bounds of this hardness score region later.

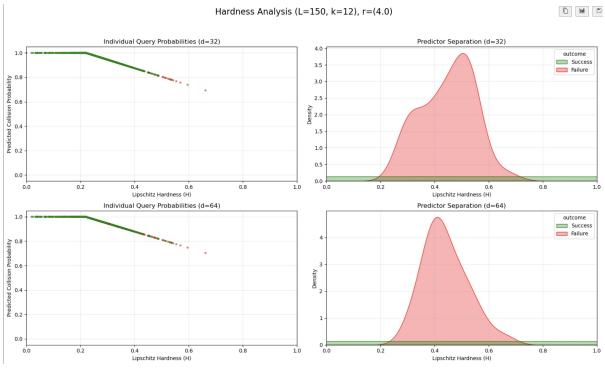
Looking at the left hand charts in **Figure 6**, we observe a distinct negative correlation: as the hardness score increases, the empirical collision probability consistently declines. This relationship confirms what we observed in **Figure 4** that our hardness metric successfully captures the geometric distortions that lead to retrieval failure. We also note that the steepness of this decline is sensitive to the LSH parameters. Specifically, the chart for the smaller bucket width ($r = 150$) exhibits a much sharper drop-off in collision probability compared to the wider bucket setting ($r = 250$). This is expected, as a tighter quantization makes the hash collisions more susceptible to perturbations caused by dimensionality reduction.



(a) GIST DJL+E2LSH at $L = 150, k = 20, r = 4$



(b) GIST DJL+E2LSH at $L = 100, k = 20, r = 4$



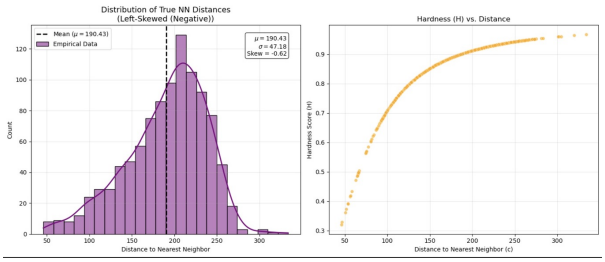
(c) GIST DJL+E2LSH at $L = 150, k = 12, r = 4$

Figure 7: **Hardness Score across different Bucket Widths in GIST** The left hand charts show collision probability vs hardness score per query where green dots are success queries and red dots are failure queries. Right hand charts show density of success or failure queries around a particular hardness score. Each chart shows different DJL dimension reductions for $d = 32$ for top charts, and $d = 64$ for bottom charts.(a) Baseline setting. (b) Fewer hash tables increases number of failure queries. (b) Fewer hash bits makes for more points per bucket increasing overall recall at the cost of complexity.

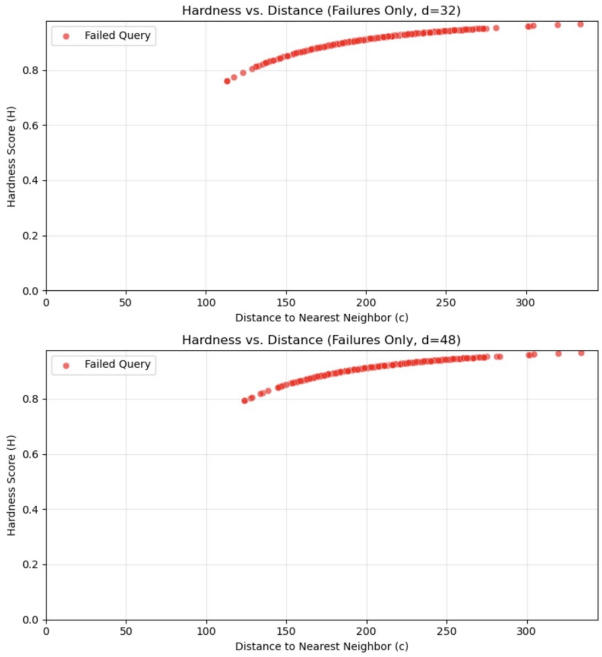
In **Figure 7** we observe how the hardness scores remain within a consistent bounded region for different number of hash tables and number of hash bits in the DJL+E2LSH algorithm on GIST. Both the right and left hand side charts show the density of the failure queries consistently from hardness scores in the range of 0.2 to 0.6. The success queries have a much broader range, hence, why there total volume in the chart is much more spread out. For the right hand charts we specifically zoomed in on the density of the failing queries to show how they consistently appear in

the same bounded region of the hardness score for different experimental settings.

Our consistent results across Figures 4 and 6 validate that our hardness function is not merely an artifact of specific hyperparameters, but a fundamental predictor of where the JL-LSH composition is likely to break down.

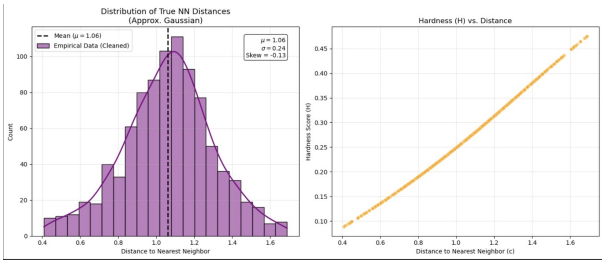


(a) SIFT DJL+E2LSH Density of neihbors at different neighbor-query distances

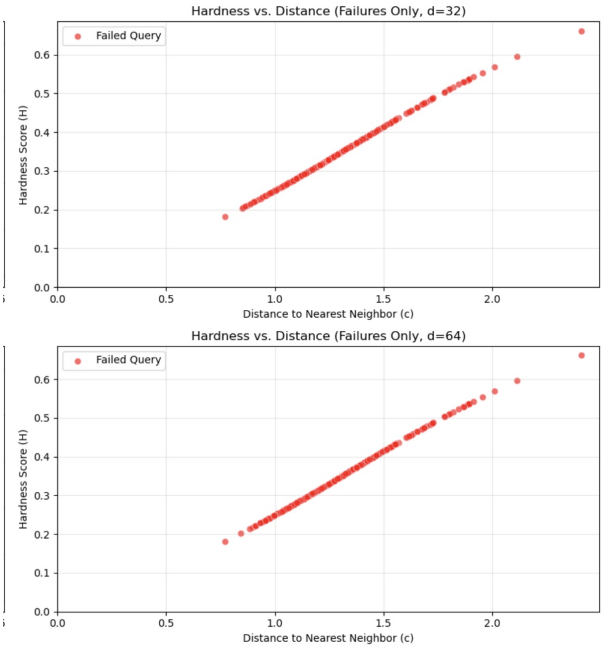


(b) SIFT DJL+E2LSH at $L = 80, k = 4, r = 250$

Figure 8: **Hardness Score vs Distance:**(a) Showing neigbor-query distances for all queries. (b) Hardness vs neighbor-query distance for only failure queries.



(a) GIST DJL+E2LSH Density of neihbors at different neighbor-query distances



(b) GIST DJL+E2LSH at $L = 150, k = 20, r = 4$

Figure 9: **Hardness Score vs Distance:**(a) Showing neigbor-query distances for all queries. (b) Hardness vs neighbor-query distance for only failure queries.

Building on our observation that failing queries are consistently bounded within specific hardness regions, we now turn to the geometric implications of these bounds. We observe distinct behavior between the two datasets:

- **SIFT Logarithmic Curve Figure 8:** The bulk of failures occur between hardness scores of 0.8 and 1.0. The relationship between Hardness and Query-Neighbor distance appears logarithmic. This nonlinear scaling likely reflects the varying local density of the SIFT manifold. SIFT descriptors, being histograms of gradients, often lie on a complex, cone-like structure. In such high-dimensional spaces, the volume of the space—and thus the potential for interfering points—grows exponentially with radius.2 Thus, the hardness score rises sharply even with small increases in neighbor distance, effectively 'saturating' near 1.0 as the query moves away from the dense cluster center.

- **GIST Linear Curve Figure 9:** In contrast, GIST failures cluster in the lower range of 0.2 to 0.6, exhibiting a linear relationship between Hardness and distance. This linearity comes from the normalized geometry of the GIST dataset.

The failure zone ($(0.8, 1.0)$ for SIFT and $(0.2, 0.6)$ for GIST) represents the geometric threshold where the distortion introduced by the JL lemma $(1 \pm \epsilon)$ exceeds the gap between the nearest neighbor and the colliding false positives.

The predictability of these failure zones validates our Hardness Function as a practical confidence indicator that can be used to set confidence values on query points for similar algorithms or as way of telling an adaptive ANN when to look deeper into a particular query.

Future Work

Throughout this paper, we introduced and analyzed the Relative Local Lipschitz Hardness metric $H_{\text{rel}}(q)$, which opens several promising directions for future work.

One immediate avenue is to use the hardness metric as an adaptive preprocessing signal. Our experimental results show that points with higher hardness scores are significantly more sensitive to distortions introduced by dimensionality reduction, leading to degraded recall. This suggests the potential for a hybrid dimensionality reduction pipeline in which queries or points with large $H_{\text{rel}}(q)$ values are projected less aggressively, while more robust points undergo stronger compression. Such an approach could preserve much of the efficiency gained from dimensionality reduction while maintaining high retrieval accuracy where it is most critical.

Relatedly, the hardness metric could serve as a confidence indicator for approximate nearest neighbor systems. Rather than treating all ANN outputs equally, a system could use $H_{\text{rel}}(q)$ to estimate the reliability of a returned neighbor. This would be particularly valuable in classification and retrieval pipelines, where incorrect neighbors can propagate downstream errors. In this way, the hardness metric transforms ANN from a black-box heuristic into a system with interpretable reliability guarantees.

Finally, while this work focuses on classical LSH families, the underlying Lipschitz-based analysis is not limited to hashing. Extending this framework to graph-based ANN methods such as HNSW, which dominate modern large-scale retrieval systems, presents a natural and impactful direction for future research.

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